



Characterization of maleic anhydride/styrene melt-grafted random copolypropylene and its impact on crystallization and mechanical properties of isotactic polypropylene

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Abstract

Maleic anhydride/styrene-grafted random copolypropylene (MPP) was prepared by multi-monomer melt grafting. Fourier transform infrared spectroscopy verified that maleic anhydride and styrene were successfully grafted onto the polypropylene random copolymer (PPR) backbones. Dynamic rheological tests indicated the existence of long branched chains on MPP. The influence of grafting degree of maleic anhydride on the crystallization behavior of MPP was investigated by differential scanning calorimetry, wide-angle X-ray diffraction and polarized optical microscopy. The results showed that the crystallization rate of MPP was higher than that of PPR, and it increased with the increase in grafting degree. The long-chain branched structure could promote heterogeneous nucleation formation of MPP, resulting in the increase in crystallization rate and the decrease in spherulite size. Moreover, the effect of MPP content on crystallization and mechanical properties of isotactic polypropylene (PP)/MPP blends was further investigated. It was found that the interfacial interaction in the blends with 1–10 wt% of MPP was higher than PP/MPP-15% blend. Introducing MPP into PP could induce the formation of β -crystal. Consequently, the incorporation of MPP into PP increased the tensile strength and impact strength and maximum values were obtained for PP/MPP-10% blend.

Keywords Random copolypropylene · Grafted · Long branched · Crystallization · Mechanical properties

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Introduction

Isotactic polypropylene (PP) is a widely used microporous membrane material owing to its relatively low cost, good mechanical properties, excellent thermal stability and chemical resistance. However, its non-polarity and low surface energy bring about the poor affinity to polar liquid electrolytes and ultimately affect the cycle performance and charge–discharge efficiency of cells [1–6]. Blending PP with an amphiphilic block or grafted copolymer such as maleic anhydride-grafted polypropylene (PP-g-MAH) has been proved to be an effective way [7–9]. On the one hand, good dispersion and strong combination can be formed between hydrophobic segments in amphiphilic copolymer and PP matrix; on the other hand, hydrophilic segments can migrate to the surface of PP products for the thermodynamic immiscibility.

Generally speaking, microporous membranes are made from a solution or phase separation process. In recent studies, a dry process was developed by combining specific crystalline morphology development in polymer film containing an oriented lamellar structure [10–13]. It should be mentioned that radical grafting of a monomer such as maleic anhydride onto PP in the presence of a peroxide initiator is often accompanied by severe degradation of PP backbone and low grafting degree of monomer [14]. Blending PP with this kind of grafted copolymer would lower the crystalline orientation factor of PP casting film and deteriorate its micropore-forming performance by uniaxial stretching [15, 16]. Our group set about the research on PP-g-MAH since 1996 [17, 18]. In order to meet the needs for the hydrophilic modification of PP microporous membrane, several measures have been taken as follows in the preparation of PP-g-MAH: Firstly, powdery polypropylene random copolymer (PPR) with low melt flow rate was used as grafting matrix resin. This is beneficial to uniform mixing between grafting agents and powdery matrix resin. Moreover, for PPR, a small amount of ethylene units distribute randomly in PP chains, in favor of restraining the degradation of PP. Secondly, styrene was used as a second grafting monomer. It has been widely reported that the addition of styrene can inhibit the degradation of PP and improve the grafting degree of maleic anhydride [7, 8, 14]. Thirdly, the amount of peroxide initiator was controlled at around 0.1 parts per hundred of resin by weight (phr) and grafting reaction temperature of twin-screw extruder around 200 °C. In our previous work, maleic anhydride/styrene melt-grafted random copolypropylene (PPR-g-(MAH/St)) was melt blended with PP at different concentrations to develop hydrophilic microporous membranes by uniaxial stretching. It was found that the hydrophilicity of the membranes was enhanced remarkably. Furthermore, the complex impedance and specific capacity of the cells with 5 wt% PPR-g-(MAH/St) also performed better, owing to its fine pore structure and electrolyte compatibility [19].

The change in macromolecular architecture will affect not only rheological property but also crystallization behavior of polymer. However, these properties of PPR-g-(MAH/St) have seldom been studied in detail compared with those of PPR. In addition, the mechanism that PPR-g-(MAH/St) impacts bulk properties

of PP is not wholly understood up to now. In this paper, PPR-*g*-(MAH/St) with different grafting degrees was prepared by multi-monomer melt grafting, and the viscoelasticity and crystallization and melting behavior were investigated to reveal the influence of grafting degree of maleic anhydride on the structure, dynamic rheological properties, as well as nucleation and crystallization behavior of PPR-*g*-(MAH/St). Moreover, the effect of PPR-*g*-(MAH/St) on intermolecular interaction, crystallization and mechanical properties of PP/PPR-*g*-(MAH/St) blends was also investigated, to provide an underlying insight needed to guide the design of PP microporous membrane.

Experimental

Materials

Polypropylene random copolymer (PPR, M_n of 109.9 kg/mol and a polydispersity of 4.5), grade R503, with ethylene content 3.2% and melt flow rate of 0.5 g/10 min (230 °C, 2.16 kg), was produced by Sinopec Yangzi Petrochemical Co. Ltd. Isotactic polypropylene (PP, M_n of 70.2 kg/mol and a polydispersity of 4.1), grade F401, with melt flow rate of 2.5 g/10 min (230 °C, 2.16 kg), was obtained from Sinopec Yangzi Petrochemical Co. Ltd. Maleic anhydride (MAH) and styrene (St) were commercially available (analytical grade). Dicumyl peroxide (DCP) used as an initiator was obtained from Fushun 3rd Chemical Factory. Antioxidant 1010 (terakis[methylene- β -(3,5-di-*tert*-butyl-4-hydroxyphenyl)-propionate] methane) was used as a stabilizer and supplied by Nanjing Milan Chemical Co. Ltd, China.

Preparation of PPR-*g*-(MAH/St)

Maleic anhydride/styrene melt-grafted random copolypropylene (PPR-*g*-(MAH/St)) was prepared in a twin-screw extruder (HT35, Nanjing Rubber and Plastics Machinery Plant Co. Ltd, China). The premix of PPR, MAH, St, DCP and antioxidant 1010, which were mixed in a high-speed mixer in advance, was fed into the extruder under the required conditions. The processing temperatures from the first zone to the ninth zone were set as 170, 180, 190, 200, 200, 200, 200, 210 and 210 °C, respectively. Three kinds of PPR-*g*-(MAH/St) prepared were denoted as MPP1, MPP2 and

Table 1 Formulation of the samples

Samples	PPR (g)	MAH (g)	DCP (g)	St (g)	Antioxidant 1010 (g)
PPR	100	0	0	0	0
MPP1	100	0.8	0.10	0.7	0.3
MPP2	100	1.2	0.12	1.0	0.3
MPP3	100	1.5	0.10	1.4	0.3

MPP3, respectively, with the same melt flow rate of 4.9 g/10 min (230 °C, 2.16 kg). The formulation of the samples is given in Table 1.

Preparation of PP/MPP blends

All materials were dried in a vacuum oven at 80 °C for 12 h prior to compounding. The proportions of MPP in PP/MPP blends were 1, 2, 5, 10 and 15% by weight, respectively. PP/MPP blends were prepared by using an internal mixer (PPT-3/LLZ-40, Guangzhou POTOP Experimental Analysis Instrument Co. Ltd, China) at 180 °C with a rotation speed of 60 rpm for 7 min. The resulting blends were then hot pressed into sheets and cut into pieces and injection molded in a micro-injection molding machine (WZS10G, Shanghai Xinshuo Precision Machinery Co. Ltd, China) for preparing the test specimens. The injection molding conditions were set as follows: melting temperature 185 °C, mold temperature 25 °C, holding pressure 0.6 MPa and holding time 10 s. To make sure that all the samples have the same thermal and mechanical history, pure PP was also processed under the same conditions.

Characterization

Fourier transform infrared spectroscopy (FTIR)

MPP samples were dissolved in refluxing xylene, and excess acetone was then added to precipitate them. Acetone could precipitate only the grafted and ungrafted PPR. The precipitated samples were filtered, washed and dried under vacuum at 80 °C for 24 h. Then, the samples were hot pressed into thin films at 180 °C for FTIR analysis. FTIR spectra were measured using a Fourier transform infrared spectroscope (Avatar 370, Nicolet, USA), which were recorded at the average of 32 scans in the standard wave number range of 500–4000 cm^{-1} at a resolution of 0.4 cm^{-1} .

Gel determination

MPP samples obtained from extruder were packed with filter paper and were then extracted in a Soxhlet extraction apparatus with xylene at 140 °C for 24 h. All the MPP samples were dissolved, and no gel was detected.

Dynamic rheological properties

Dynamic oscillatory rheological measurements of pure PPR, MPP, PP and PP/MPP blends were taken with a rotary rheometer (Physica MCR 301, Anton-Paar, Austria), with parallel plate geometry of diameter 25 mm and gap 1 mm, and a nitrogen atmosphere was used to avoid thermal degradation. The complex viscosity and storage modulus as a function of angular frequency were determined within the linear viscoelastic regime with a frequency range from 0.01 to 628 rad/s. The weighted

relaxation spectra were evaluated from the dynamic moduli data using nonlinear regularization software.

Wide-angle X-ray diffraction (WAXD)

The crystalline structure was investigated using a wide-angle X-ray diffractometer (D/MAX2500, Rigaku, Japan), employing Cu K α radiation ($\lambda = 1.54 \text{ \AA}$), and the measurements were taken with a 2θ range of 5° – 35° at a scanning rate of $2^\circ/\text{min}$. The core region of the injection molded sample was characterized so that the crystalline structure of different samples could be well compared.

Differential scanning calorimetry (DSC)

The crystallization and melting behaviors were investigated using a differential scanning calorimeter (DSC 6000, PerkinElmer, USA) in a nitrogen atmosphere. Samples of about 8 mg were heated from 30 to 210°C at a heating rate of $10^\circ\text{C}/\text{min}$ and maintained at 210°C for 5 min to eliminate thermal history. Then, the samples were cooled to 30°C at a cooling rate of $10^\circ\text{C}/\text{min}$, following a second heating from 30 to 210°C at a rate of $10^\circ\text{C}/\text{min}$. The first cooling and second heating scans were recorded. The procedures for isothermal crystallization were set as follows: The PPR and MPP samples were melted at 210°C for 5 min to eliminate thermal history. Then, they were rapidly cooled to the crystallization temperature and maintained the necessary time.

Polarized optical microscopy (POM)

The crystal morphology of PPR and MPP samples was observed with a polarized optical microscope (ECLIPSE 50i POL, Nikon, Japan). The samples were hot pressed into thin films at 200°C on the hot stage and held for 5 min to eliminate thermal history. They were then quenched to 115°C and maintained for 40 min to allow crystallization. Finally, the films were quickly cooled to room temperature for the observation of crystal morphology.

Mechanical properties measurements

Tensile measurements were taken on a universal tensile testing machine (WDT-5, Shenzhen Kaiqiang Experimental Instrument Co. Ltd, China) according to GB/T 1040-2006 at a tensile speed of $50 \text{ mm}/\text{min}$. The Izod notched impact strength of the specimens was measured with an impact tester (XJU-22, Chengde Testing Machine Co. Ltd, China) according to GB/T 1843-2008. The values of all the mechanical properties were calculated as averages over five specimens. All the measurements were taken at room temperature (25°C).

Fig. 1 FTIR spectra of PPR and MPP samples

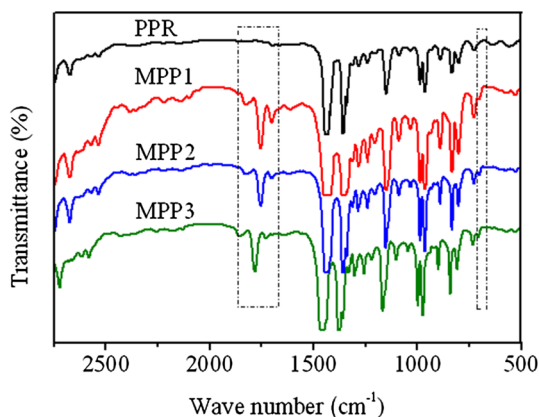


Table 2 Relative grafting degrees of maleic anhydride and styrene

Samples	PPR	MPP1	MPP2	MPP3
R_{MAH}	0	1.38	2.25	2.42
R_{St}	0	0.10	0.11	0.12

Scanning electron microscopy (SEM)

Impact-fractured surfaces were characterized using a field emission scanning electron microscope (SUPRA 55, Zeiss, Germany), with an accelerating voltage of 5 kV. All the specimens were sputter-coated with a thin layer of gold prior to SEM characterizations.

Results and discussion

Characterization of MPP

Long-chain branched structure

The FTIR spectra of PPR and MPP are shown in Fig. 1. In the case of PPR, the absorption bands of PP at 997 and 973 cm^{-1} and absorption band of methylene groups at 731 cm^{-1} can be seen, which is ascribed to single ethylene chain link inserting into PP. No absorption band appears at 720 cm^{-1} , indicating that PPR we used is polypropylene random copolymer with ethylene chain links inserting singly rather than continuously into PP [20, 21]. As for MPP, besides the characteristic absorptions of the PPR skeleton, there are several new bands. The bands at 1857, 1782 and 1728 cm^{-1} can be ascribed to anhydride groups, and band at 706 cm^{-1} can be assigned to phenyl groups, implying that maleic anhydride and styrene were successfully grafted onto the PPR backbones.

Fig. 2 Complex viscosity (η^*) and storage modulus (G') as a function of angular frequency for PPR and MPP melts at 160 °C

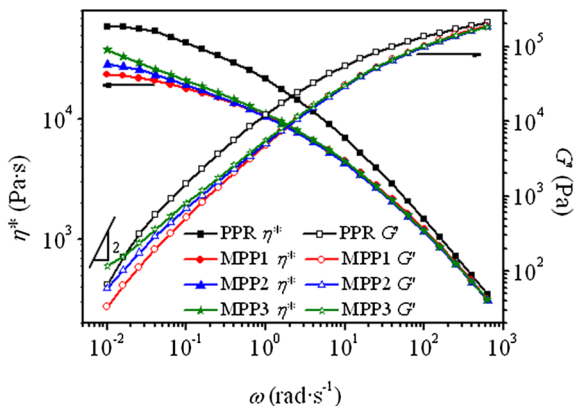
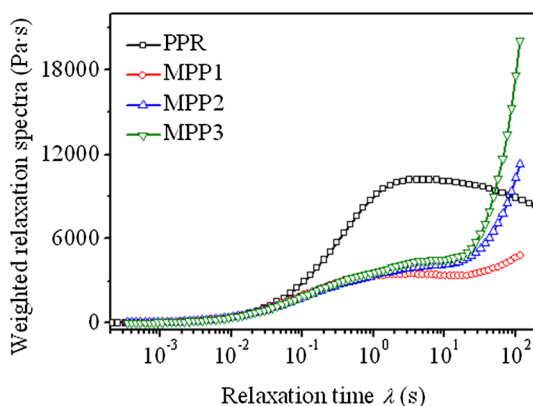


Fig. 3 Weighted relaxation spectra for PPR and MPP samples



From the FTIR spectra, the relative grafting degrees of maleic anhydride and styrene (expressed in R_{MAH} and R_{St}) are defined as A_{1782}/A_{810} and A_{706}/A_{810} , respectively, where A_{1782} is the band area at 1782 cm^{-1} , corresponding to the absorption of carbonyl groups in grafted maleic anhydride, A_{706} is the band area at 706 cm^{-1} , characteristic of the absorption of grafted styrene and A_{810} is the band area at 810 cm^{-1} , corresponding to the absorption of methyl groups used as internal standard [22]. Values of R_{MAH} and R_{St} are listed in Table 2. It can be seen that the R_{St} values are close to each other and relatively lower compared with the R_{MAH} values. Therefore, R_{MAH} is used to characterize the total grafting degree of MPP in this paper. The grafting degree of these MPP samples follows the order of $\text{MPP1} < \text{MPP2} < \text{MPP3}$.

Figure 2 displays the curves of complex viscosity (η^*) and storage modulus (G') as a function of angular frequency (ω) for PPR and MPP melts at 160 °C. Typical shear thinning behavior can be observed. The η^* values of MPP are lower than those of PPR at the same frequency. This can be explained by the degradation of PPR in the process of melt grafting. However, due to the same melt flow rate, a coincidence of η^* - ω curves at high frequencies for these MPP samples can be observed.

Linear PPR exhibits typical terminal behavior, with the slope of G' - ω curve at low-frequency region close to 2 [23]. However, all the MPP samples show a

deviation from the terminal behavior. With the increase in grafting degree, the terminal slope decreases gradually, implying the existence of a longer relaxation time. In non-filled systems, this can be attributed either to cross-linking or to branching [24, 25]. Considering the fact that no gel was detected in all the MPP samples, the non-terminal effect could be attributed to the long branched chains produced by melt grafting reaction [26, 27].

The weighted relaxation spectra of PPR and MPP samples were further investigated to understand their relaxation mechanism, as indicated in Fig. 3. It is obvious that there is only one characteristic relaxation time for PPR, related to the relaxation of linear macromolecule. However, two characteristic relaxation times can be observed for MPP samples, the smaller one corresponding to the linear backbones and the longer one related to the branched chains. With the increase in grafting degree of MPP, the longer relaxation time increased and exceeded the maximum relaxation time that could be determined from our experiments. The results of dynamic rheological tests further verify the existence of long branched chains on MPP.

According to the investigation by Xie et al., two types of grafting process may occur in the case of maleic anhydride and styrene dimonomer melt grafted onto PP. (1) Styrene reacted with maleic anhydride and styrene–maleic anhydride copolymer (SMA) was produced by free radical reaction, and then SMA grafted onto PP to form branched chains. (2) PP was initiated to generate macroradicals, and then styrene grafted onto PP chains to form more stable styryl macroradicals, which could suppress the degradation of PP. Finally, maleic anhydride grafted to the stable macroradicals to increase the length of branched chains [14]. Therefore, with the increase in grafting degree of MPP, the branched chains became longer, resulting in the increase in complex viscosity at low-frequency region as well as longer characteristic relaxation time.

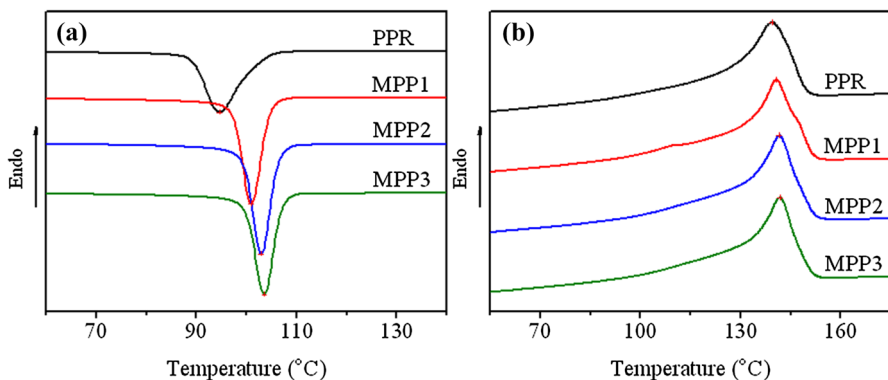


Fig. 4 Crystallization (a) and melting curves (b) for PPR and MPP samples

Table 3 Crystallization and melting parameters obtained from the DSC thermograms for PPR and MPP samples

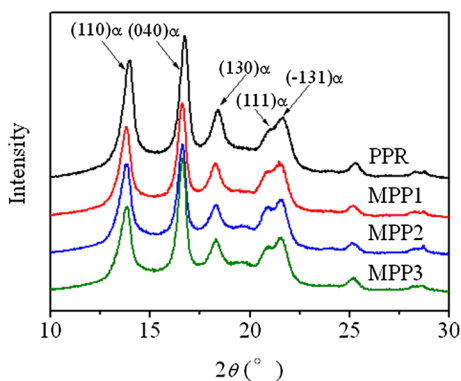
Samples	T_c (°C)	T_m (°C)	ΔH_m (J g ⁻¹)
PPR	94.69	139.41	65.96
MPP1	100.93	140.77	69.08
MPP2	103.01	141.58	74.70
MPP3	103.62	141.91	68.99

Crystallization and melting behavior

The change in macromolecular architecture could affect not only rheological property but also crystallization behavior of PPR. Figure 4(a) and (b) presents the crystallization and melting curves of PPR and MPP samples, respectively. The corresponding parameters obtained from the DSC thermograms are listed in Table 3. It is found that the values of crystallization temperature (T_c), melting temperature (T_m) and enthalpy of fusion (ΔH_m) of MPP are higher than those of PPR. The crystallization peaks of MPP samples shift about 6–9 °C toward higher temperature. This can be attributed to the heterogeneous nucleation effect of long branched chains on MPP. Similar phenomena were observed by other researchers who reported a nucleation effect of carbonyl groups on the crystallization of PP [28, 29]. The higher the grafting degree, the more remarkable the heterogeneous nucleation. Furthermore, degradation of PPR in the process of melt grafting brought out lower melt viscosity of MPP, which facilitated the diffusion and regular arrangement of MPP chains toward crystal nuclei, resulting in the increase in T_m and ΔH_m values.

Crystalline structure

Figure 5 presents the WAXD profiles of PPR and MPP samples. It can be observed that the WAXD profile of PPR shows five obvious peaks at 2θ of approximate values 14.0°, 16.7°, 18.4°, 21.0° and 21.7°, respectively, which correspond to the (110), (040), (130), (111) and (−131) reflections, indicating that PPR can form only

Fig. 5 WAXD patterns for PPR and MPP samples

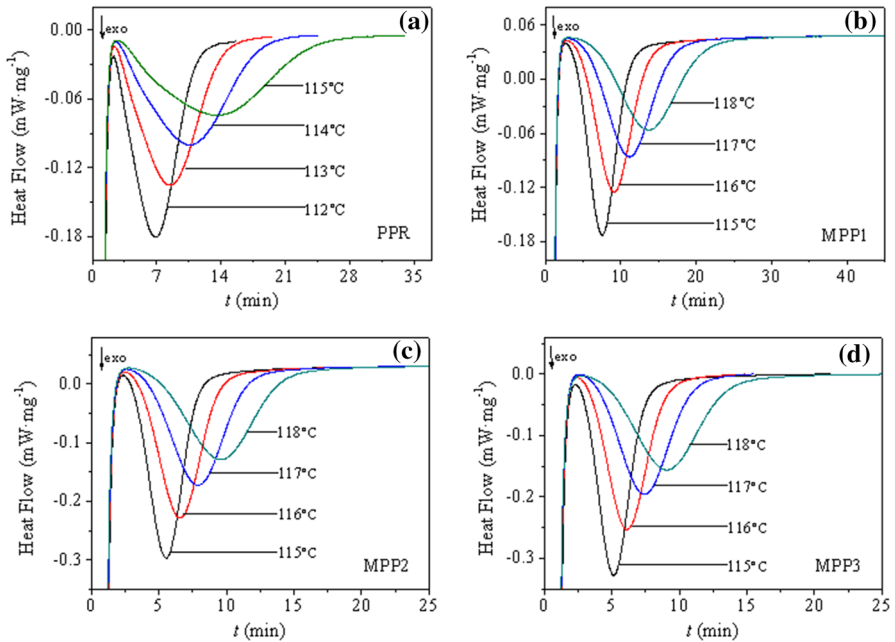


Fig. 6 DSC curves of isothermal crystallization for PPR and MPP

α -crystal. For MPP samples, there is no new peak in contrast to PPR, indicating that the crystal form of MPP is the same as that of PPR.

Isothermal crystallization kinetics

In order to understand the effect of long branched chain structure on crystallization rate of MPP, the isothermal crystallization kinetics of PPR and MPP samples were further investigated. The DSC curves of isothermal crystallization for PPR and MPP are presented in Fig. 6. According to Avrami kinetics, the change in relative crystallinity $X(t)$ with time t during isothermal crystallization can be described by:

$$\lg [-\ln(1 - X(t))] = \lg K + n \lg t \quad (1)$$

where n is the Avrami exponent and K is the crystallization rate constant. The plots of $\lg[-\ln(1 - X(t))]$ versus $\lg t$ show good linear portion, as presented in Fig. 7. The slope of linear fitting represents the Avrami exponent n , and the intercept is $\lg K$. The half crystallization time ($t_{1/2}$) can be obtained when the relative crystallinity $X(t)$ is equal to 0.5. Table 4 shows the list of the parameters of isothermal crystallization at different temperatures obtained from Fig. 7. It can be seen that in the temperature range we studied, the n values of PPR range from 2.2 to 2.5 and the n values of MPP are greater than 3. At the same crystallization temperature (115 °C), the n value of MPP is significantly higher, indicating that the number of MPP crystal nuclei is more than that of PPR [30], and the long branched chains grafted onto

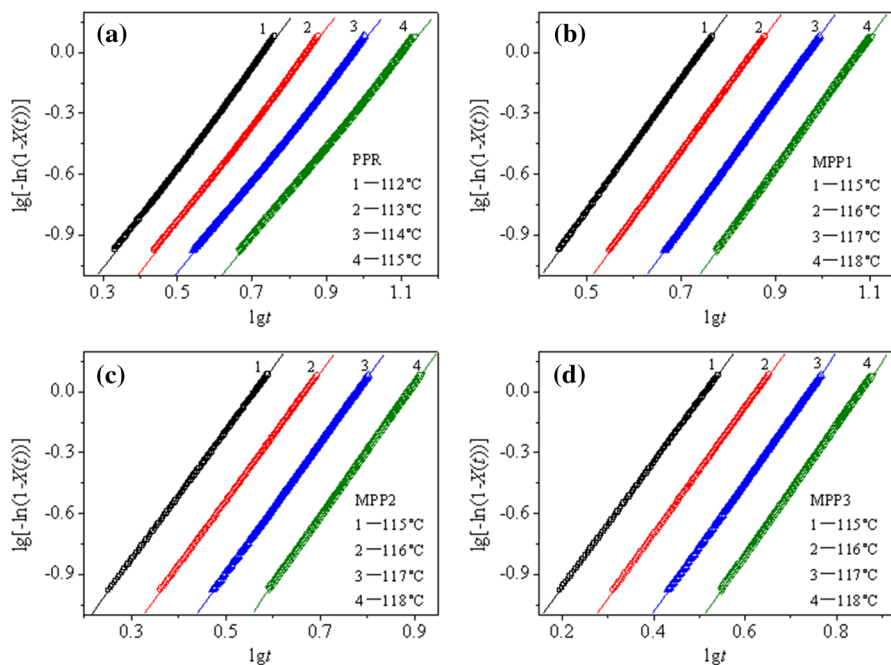


Fig. 7 Plots of $\lg[-\ln(1-X(t))]$ versus $\lg t$ for PPR and MPP at different temperatures

Table 4 Kinetic parameters of isothermal crystallization for PPR and MPP

Samples	T_c (°C)	n	K (min $^{-n}$)	$t_{1/2}$ (min)
PPR	112	2.46	0.0157	4.66
	113	2.40	0.00905	6.08
	114	2.30	0.00574	8.01
	115	2.25	0.00321	10.85
MPP1	115	3.18	0.00450	4.87
	116	3.23	0.00188	6.23
	117	3.24	0.000782	8.12
	118	3.41	0.000220	10.62
MPP2	115	3.09	0.0200	3.15
	116	3.15	0.00818	4.09
	117	3.15	0.00384	5.20
	118	3.25	0.00142	6.72
MPP3	115	3.04	0.0277	2.88
	116	3.12	0.0111	3.76
	117	3.09	0.00555	4.77
	118	3.19	0.00194	6.31

PPR play the role of heterogeneous nucleation. Comparing the values of $t_{1/2}$ and K for PPR and MPP at the same crystallization temperature, it can be concluded that the crystallization of MPP is obviously accelerated. The study of Seo et al. demonstrates that introducing grafted monomers may result in the reduction in the regularity of macromolecular chains and the enhancement of steric hindrance for folding of propylene sequences. On the other hand, branched chains can also act as a nucleating agent for the crystallization of propylene sequences [31]. It could be seen that with the increase in crystallization temperature, the $t_{1/2}$ value increased and the K value decreased, indicating that the temperatures we chose are in the high-temperature region for PPR and MPP crystallization, which is dominated by nucleation. The existence of long branched chains promotes the heterogeneous nucleation of MPP, resulting in a remarkable increase in crystallization rate. With the grafting degree increasing, $t_{1/2}$ decreased and K value increased gradually, implying the introduced branching structures speeded up the crystallization.

The crystal morphology of PPR and MPP samples was observed by polarized optical microscope (POM), as displayed in Fig. 8. As shown in Fig. 8a, the spherulites of linear PPR are larger, with obvious boundaries and non-uniform size distribution. However, for MPP as shown in Fig. 8b–d, a large number of finer spherulites can be observed in contrast to PPR. This can be explained by the heterogeneous nucleation of long branched chains. Plenty of crystal nuclei formed and grew at the same time, resulting in a remarkable decrease in spherulite size. The higher the grafting degree, the smaller the spherulite size.

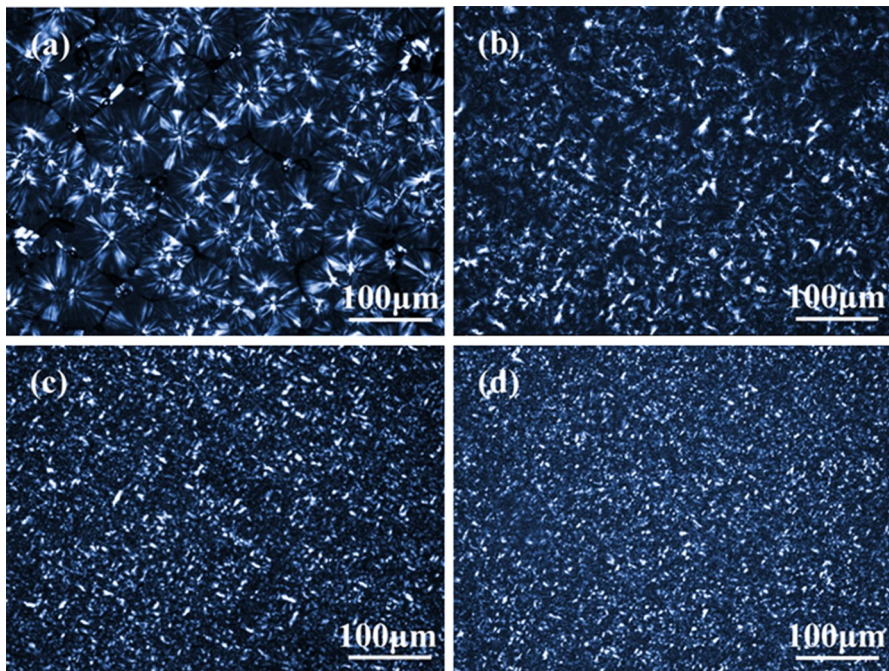


Fig. 8 Polarized optical micrographs of **a** PP, **b** MPP1, **c** MPP2 and **d** MPP3

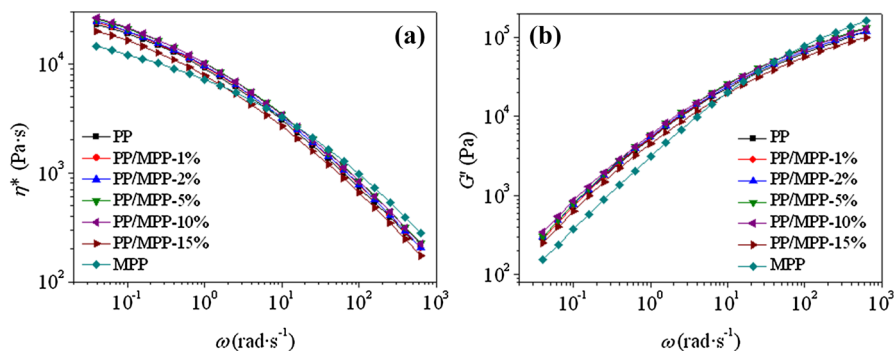


Fig. 9 Complex viscosity (a) and storage modulus (b) of PP, MPP and PP/MPP blends as a function of angular frequency at 180 °C

Dynamic rheological behavior of PP blended with MPP

To investigate the intermolecular interaction between PP and MPP, dynamic oscillatory rheological measurements for PP/MPP blends were taken. MPP2 is taken for an example and denoted as MPP in the following sections. Figure 9a, b presents the results of complex viscosity and storage modulus obtained for pure PP, MPP and PP/MPP blends as a function of angular frequency at 180 °C, respectively. All the samples exhibited typical shear thinning behavior. It is interesting to note that the addition of 1–10 wt% of MPP into PP increases the viscosity and elasticity at low frequencies, and the PP/MPP-15% blend possessed intermediate values between pure PP and MPP.

Due to the similar chemical structure of PP and MPP as well as the low content of maleic anhydride groups in MPP, it is very difficult to characterize the phase behavior of PP/MPP blends. Results of linear viscoelastic tests applied in small amplitudes can provide reliable information on the microstructure of the blends. Particularly, the viscoelastic response at low-frequency region can be used for evaluating the interfacial interaction between phases [32]. The complex viscosity and storage

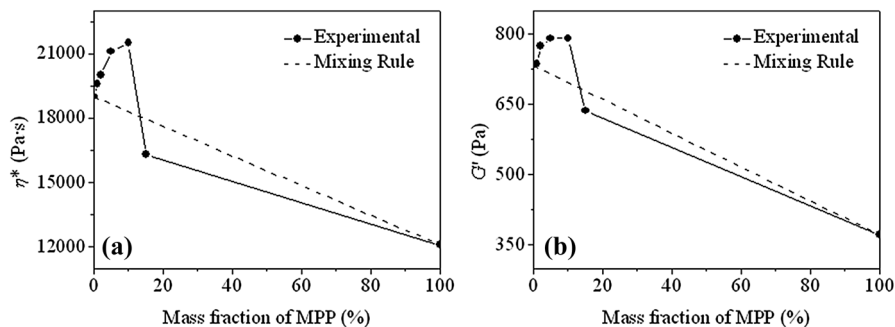


Fig. 10 Composition evolution of complex viscosity (a) and storage modulus (b) of PP/MPP blends at an angular frequency of 0.1 rad/s

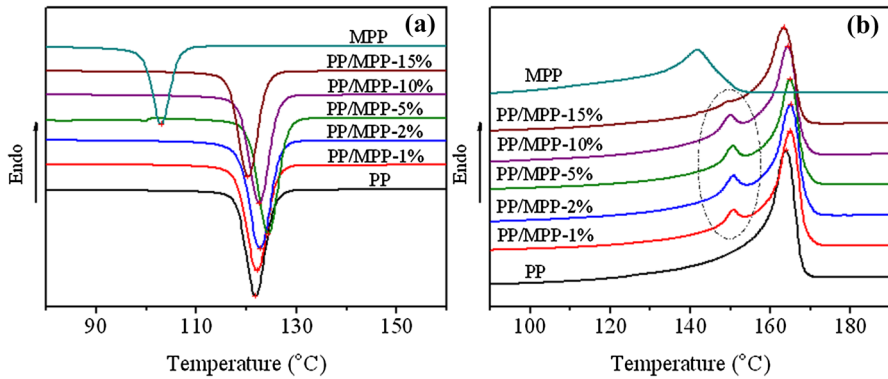


Fig. 11 DSC curves of PP, MPP and PP/MPP blends on the first cooling (a) and the second heating scan (b)

Table 5 Crystallization and melting parameters of PP, MPP and PP/MPP blends obtained from the DSC thermograms

Samples	T_c (°C)	T_m (°C)	ΔH_m (J g ⁻¹)	X_c (%)
MPP	103.01	141.58	74.70	36.09
PP	121.76	163.89	108.31	52.32
PP/MPP-1%	122.11	164.96	114.02	55.08
PP/MPP-2%	122.72	165.06	120.49	58.21
PP/MPP-5%	124.40	164.81	126.86	61.29
PP/MPP-10%	122.60	164.29	115.35	55.72
PP/MPP-15%	120.26	163.31	98.58	47.62

modulus versus MPP mass fraction together with the results calculated using linear mixing rule at an angular frequency of 0.1 rad/s are shown in Fig. 10a, b, respectively. It can be seen obviously that the complex viscosity and storage modulus show a remarkable positive deviation from mixing rule for the blends with 1–10 wt% of MPP, while they show negative deviation for PP/MPP-15% blend. On the basis of Utracki's research [33], this typical positive negative deviation behavior (PNDB) is usually observed for the blends in which the interfacial interaction between phases is affected by the blend composition. In other words, from the dynamic rheological results it can be concluded that for the blends with 1–10 wt% of MPP, the presence of strong interfacial interaction increases the viscosity and elasticity. On the other hand, negative deviation of the viscosity and elasticity observed for PP/MPP-15% blend can be ascribed to the presence of weak interfacial interaction between phases.

Crystallization behavior of PP blended with MPP

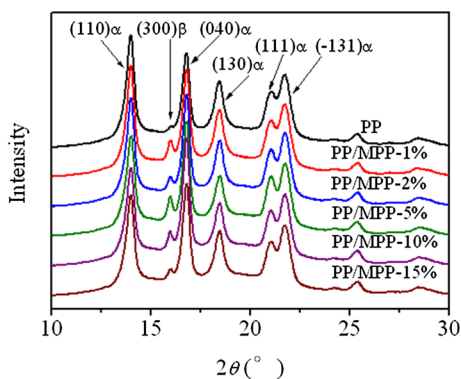
Figure 11a, b shows the DSC curves of PP, MPP and PP/MPP blends on the first cooling and the second heating scan, respectively. The corresponding crystallization and melting parameters are listed in Table 5. It can be observed that introducing

1–10 wt% of MPP into PP causes a slight increase in T_c , T_m and crystallinity (X_c). When the mass fraction of MPP is 5%, the value of T_c for the blend is increased by 2.64 °C compared with that of PP, indicating that the PP component in this blend generated more perfect crystals. Nitta et al. investigated the miscibility for the PP/PPR binary blends and summarized that PPR with a propylene content of more than 84 mol% was miscible with PP [34]. Cho et al. studied the crystallization and melting behavior of PP/PPR blends, and *co*-crystallization phenomenon was observed [35]. As mentioned above, there are maleic anhydride and phenyl groups in branched chains of MPP, which could serve as heterogeneous nucleating agent. Therefore, in the process of PP *co*-crystallizing with PPR backbones on MPP, branched chains on MPP could also act as crystal nuclei. Moreover, the viscosity of MPP melt was lower compared with that of PP, which favored the diffusion and orderly arrangement of PP chains toward crystal nuclei. However, as the concentration of MPP increases to 15%, phase separation may occur, resulting in unfavorable interfacial interaction as mentioned in dynamic rheological measurements, followed by the decline in the values of T_c and X_c .

From Fig. 11b, it can be seen that the melting curve of PP only displays a single melting peak at 163.89 °C to α -crystal, while the melting curves of PP/MPP blends show another melting peak at around 150 °C, which is caused by β -crystal, implying that introducing a certain amount of MPP can induce the formation of β -crystal in PP. According to the investigation by Su et al., it is speculated that the benzene rings of MPP chains may act as a growth surface of crystal and induce helix chains of PP epitaxial crystallize on it [36].

Figure 12 presents the WAXD profiles of PP and PP/MPP blends. It can be observed that the WAXD profile of PP shows five obvious peaks at 2θ of approximate values 14.1°, 16.9°, 18.5°, 21.1° and 21.7°, respectively, which correspond to the (110), (040), (130), (111) and (-131) reflections, indicating that pure PP can form only α -crystal. However, there is an obvious peak at about 15.9° for PP/MPP blends, which corresponds to the (300) reflection of β -crystal form, implying that β -crystal form exists in these blends. The intensity of β -crystal diffraction

Fig. 12 XRD patterns of PP, MPP and PP/MPP blends



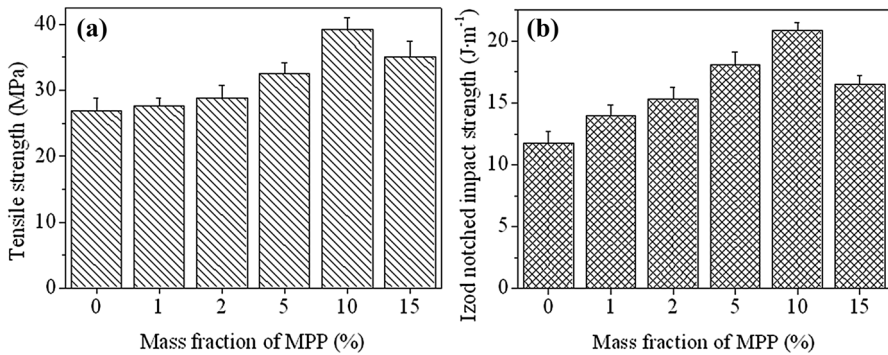


Fig. 13 Effect of mass fraction of MPP on tensile strength (a) and impact strength (b) of PP/MPP blends

peak changed little with the increase in MPP concentration. Only the intensity of α -crystal diffraction peak at 2θ value 16.9° increased slightly, implying that MPP makes PP present preferential growth in (040) crystal plane.

Mechanical properties of PP blended with MPP

Figure 13a, b presents the tensile strength and impact strength of PP/MPP blends with respect to the mass fraction of MPP, respectively. It could be seen that the incorporation of MPP into PP increased the tensile strength and impact strength and maximum value was obtained for PP/MPP-10% blend, which was followed by a decrease. When the mass fraction of MPP was 10%, the values of tensile strength and impact strength of the blends were 39.28 MPa and 20.82 J m⁻¹, which increased by 46.13 and 77.04%, respectively, compared with that of PP. As mentioned earlier, the increase in mass fraction of MPP increased the interfacial interaction between phases. Moreover, a small amount of β -crystals formed could also facilitate the improvement in PP toughness. However, beyond 10 wt% of MPP in the blend, unfavorable interfacial interaction between phases brought about the decrease in the mechanical properties of the blend.

For better understand the change in mechanical properties of PP/MPP samples, the impact-fractured surface morphology of PP and representative PP/MPP-10% samples was further characterized by using SEM, and the results are shown in Fig. 14. At a low magnification as shown in Fig. 14a, b, it can be seen that the impact-fractured surface of PP/MPP-10% blend is rougher than that of PP, indicating more energy absorption for the PP/MPP-10% blend during the impact fracture process. More differences in surface morphology can be clearly seen from the SEM micrographs obtained at higher magnification. Generally speaking, the fracture of a sample under the load condition includes two stages, i.e., crack initiation process (zone 1 as indicated in Fig. 14a1, b1) and crack propagation process (zones 2 as indicated in Fig. 14a2, b2) [37, 38]. Hence, the morphology in different zones of the fractured surface was carefully characterized at higher magnification. For the PP sample, the impact-fractured surface in the crack initiation zone is rather smooth. However, the incorporation of MPP induces much different surface morphology in

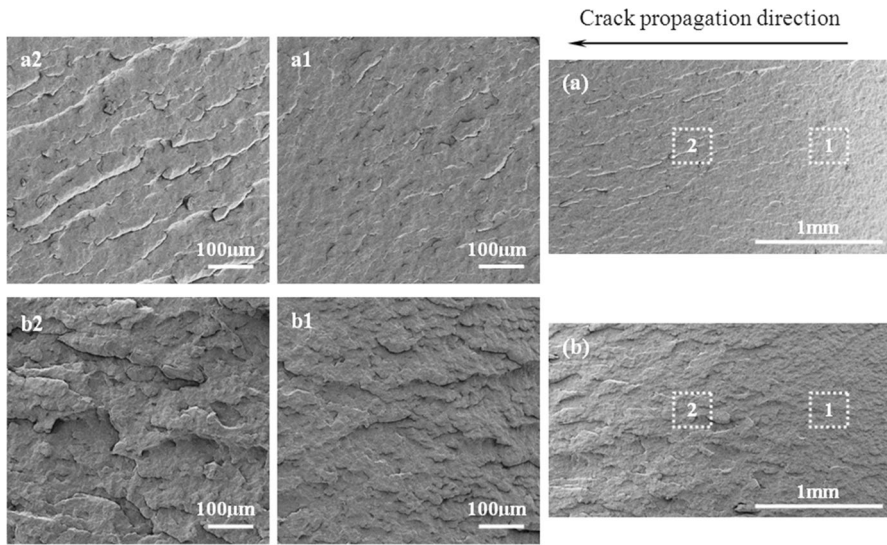


Fig. 14 SEM micrographs showing the impact-fractured surface morphology of **a** PP and **b** PP/MPP-10% blend. **a, b** Micrographs were obtained at low magnification, and the scale bar is 1 mm. **a1–b2** Micrographs were obtained from the different zones of the impact-fractured surface as indicated in micrographs **a** and **b**, and they show the morphological characteristics of samples at high magnification. The scale bar is 100 μm . The arrow shows the impact fracture direction

the PP/MPP-10% blend. The impact-fractured surfaces in the both crack initiation zone and crack propagation zone are rougher than those of PP. Therefore, it can be deduced that there is a certain relationship between the enhanced impact strength and the intensified plastic deformation of the PP blended with MPP. Introducing MPP into PP facilitates the improvement in PP impact toughness.

Conclusions

In the current study, maleic anhydride/styrene-grafted random copolypolypropylene (MPP) was prepared by melt grafting polymerization using maleic anhydride as the grafting monomer and styrene as the comonomer. Fourier transform infrared spectroscopy (FTIR) verified that maleic anhydride and styrene were successfully grafted onto the polypropylene random copolymer (PPR) backbones. Investigation on dynamic rheological properties of MPP indicated the existence of long-chain branched structure. Isothermal crystallization behavior of PPR and MPP was measured using differential scanning calorimetry (DSC), and the kinetics of isothermal crystallization was analyzed with Avrami equation. Effect of grafting degree of maleic anhydride on the kinetics of isothermal crystallization of MPP was examined. The results showed that MPP crystallization rate was higher than that of PPR, and it increased with the increase in grafting degree. The long-chain branched structure could promote heterogeneous nucleation formation of MPP, resulting in the increase in crystallization rate. The results of polarized light microscopy (POM)

verified that the long-chain branched structure could promote heterogeneous nucleation and decrease crystal size.

Moreover, the effect of MPP content on crystallization and mechanical properties of PP/MPP blends was further investigated. It was found that the increase in mass fraction of MPP increased the interfacial interaction between phases. In addition, introducing MPP into PP could induce the formation of β -crystal, which also facilitated the improvement in PP toughness. However, beyond 10 wt% of MPP in the blend, unfavorable interfacial interaction between phases brought about the decrease in the mechanical properties of the blend. Anyhow, this work provides guidance for the preparation of hydrophilic PP microporous membrane as well as better understanding of the compatibilization in PP composites with an amphiphilic grafted copolymer.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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